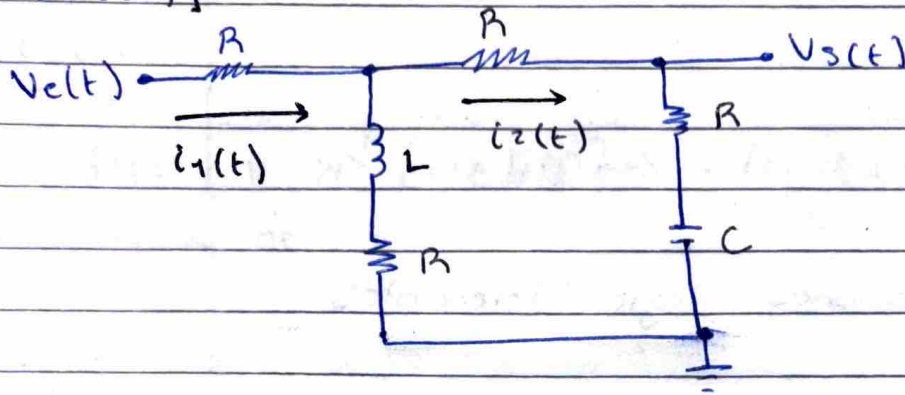


5.1 Práctica: Diseño de controlador para un sistema de 2^{do} Orden

Obj. Diseñar controlador que permita eliminar el error entre entrada $[V_e(t)]$ y salida $[V_s(t)]$



Ecuaciones principales:

Volt: entrada

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

mallo 2

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$\rightarrow V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$\rightarrow L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$\rightarrow V_s(s) = R I_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico

función de transferencia

$$\hookrightarrow \frac{V_o(s)}{V_e(s)} = \frac{(\text{polinomio}(s)) I_2(s)}{(s^2 + s + ?)(?)}$$

- ① Se deben agrupar corrientes $I_1(s)$ e $I_2(s)$
- ② Expresar el $V_e(s)$ en función de $I_2(s)$
- ③ Despejar $I_1(s)$ de la segunda ec. principal

$$\textcircled{1} \quad V_o(s) = (R + \frac{1}{Cs}) I_2(s) = \left(\frac{CRs + 1}{Cs} \right) I_2(s)$$

$$\bullet L s I_1(s) - L s I_2(s) + R I_1(s) - R I_2(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$\bullet L s I_1(s) + R I_1(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs} + L s I_2(s) + R I_2(s)$$

$$\bullet (Ls + R) I_1(s) = \left(Ls + \frac{1}{Cs} + 3R \right) I_2(s)$$

$$\bullet I_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R \right) I_2(s)$$

$$I_1(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s)$$

$$\textcircled{2} \quad V_e(s) = R I_1(s) + L s I_1(s) - L s I_2(s) + R I_1(s) - R I_2(s)$$

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s) - (Ls + R) I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} \right] I_2(s)$$

$$V_e(s) = \left[\frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs} \right] I_2(s)$$

$$\frac{V_o(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs} I_2(s)}{I_2(s)} = \frac{CRs + 1}{Cs} \frac{(Ls + 2R)(CLs^2 + 3CRs + 1) - (Ls + R)(CLs^2 + CRs)}{CLs^2 + CRs}$$

$$V_e(s) = \left[\frac{3CR^2Ls^2 + 5CR^2s + 2R + Ls}{CLs^2 + CRs} \right] I_2(s)$$

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$$V_e(s) = \left[\frac{3CLR s^2 + 5CR^2 s + 2R + Ls}{(Ls^2 + CRs)} \right] I_2(s)$$

$$V_s(s) = \frac{CRs + 1}{Cs} I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \left[\frac{3CLR s^2 + 5CR^2 s + 2R + Ls}{(Ls^2 + CRs)} \right] I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLR s^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R}$$

Polinomio de 2º Orden

Error en estado estacionario cuando se aplica un escalón unitario de entrada

$$e(s) = \lim_{s \rightarrow 0} s \left(1 - \frac{CLR s^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R} \right) = \frac{R}{2R} = 0.5V$$

Estabilidad en lazo abierto

Se calculan los polos del sistema (raíces del denominador)

$$R = 10k \quad L = 680\mu H \quad C = 100\mu F$$

$$s_1 = -0.4$$

$$s_2 = -2.45 \times 10^7$$

$$3CLR s^2 + (5CR^2 + L)s + 2R = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sist. estable

$$\lambda_1 = -4.0$$

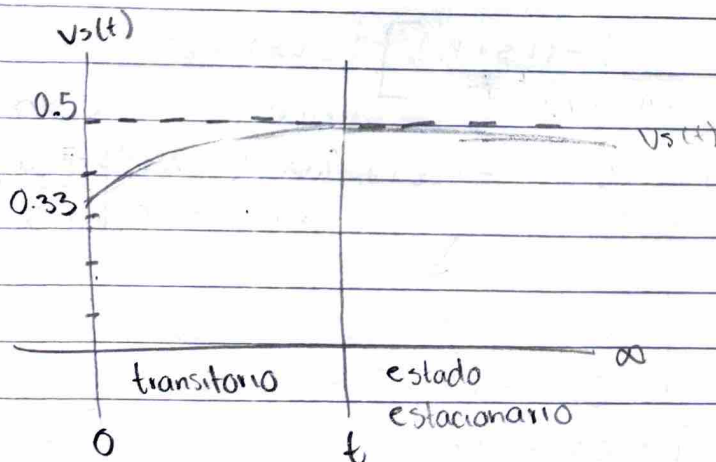
$$\lambda_2 = -1.6667 \times 10^6$$

$$a = 20.4 \times 10^{-3}$$

$$b = 50 \times 10^3$$

$$c = 20k$$

Respuesta subamortiguada



Modelo de ecuaciones integra-diferenciales

A partir de las ecuaciones principales del sistema, se despejan las variables dependientes, es decir, las corrientes y el voltaje de salida:

$$i_1(t) = \left[V_e(t) + R i_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + R i_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLBs^2 + (CB^2 + L)s + R}{3CLBs^2 + (5CB^2 + L)s + 2R}$$

$$R = 100 \Omega$$

$$L = 680 \mu H$$

$$C = 100 \mu F$$

$$\frac{V_s(s)}{V_e(s)} = \frac{6.8 \times 10^{-8} s^2 + 1.00068s + 100}{2.04 \times 10^{-5} s^2 + 5.0004 \times 10^{-3} s + 200}$$